fn make_fully_adaptive_composition

Michael Shoemate

This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `fn make_fully_adaptive_composition`.

1 Hoare Triple

Precondition

Compiler-verified

- Argument `input_domain` of type `DI`.
- Argument `input_metric` of type `MI`.
- Argument `output_measure` of type `MO`.
- Generic `DI` implements `Domain`.
- Generic `MI` implements `Metric`.
- Generic `MO` implements `Measure`.
- `(DI, MI)` implements `MetricSpace`.

User-verified

Pseudocode

```
def make_sequential_odometer(
    input_domain: DI,
    input_metric: MI,
    output_measure: MO,
) -> Odometer[DI, MI, MO, Measurement[DI, TO, MI, MO], TO]:
    def function(arg: DI_Carrier, wrapper: Wrapper | None):
        return new_sequential_odometer_queryable(
            input_domain,
            input_metric,
            output_measure,
            arg,
            wrapper)

    return Odometer.new(
        input_domain,
        Function.new_interactive(function),
        input_metric,
        output_measure)
```

Postcondition

For every setting of the input parameters `(input_domain, input_metric, output_measure, DI, TO, MI, MO)` to `make_fully_adaptive_composition` such that the given preconditions hold, `make_fully_adaptive_composition` raises an exception (at compile time or run time) or returns a valid odometer. A valid odometer has the following properties:

1. (Data-indepdendent exceptions). For every pair of elements x, x' in `input_domain`, `function`(x) and `function`(x') either both raise an exception, or neither raise an exception.

2. (Valid odometer queryable). For every element x in `input_domain`, where `function`(x) does not raise an exception, `function`(x) returns a valid odometer queryable.

Proof. (Data-independent exceptions). The only function called, `new_sequential_odometer_queryable`, does not raise an exception, as verified by the compiler. Therefore all invocations of `function` do not raise an exception. □

Proof. (Valid odometer queryable). Under the assumption that the input data is a member of the input domain, the precondition of `make_fully_adaptive_composition_queryable` is met, so by its postcondition the return value is a valid odometer queryable. □